LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 3 - trial 3 (seed 1240) - scaffold splitting

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN model on a compound activity prediction dataset, split the data using scaffold splitting. Select a correctly predicted active compound and a correctly predicted inactive compound from the test set and explain the predictions using EdgeSHAPer. Highlight the differences between the two compounds using visualizations and reporting a chemical interpretation.

Assistant

I'll help you train a GNN model on a compound activity prediction dataset, perform scaffold splitting, and explain predictions using EdgeSHAPer. Let me start by exploring the available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the binary activity prediction dataset and start the workflow. 🚀

(Calling tool `load_dataset`)

Assistant

Let me use the correct label column name:

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset is loaded with 8,348 samples. Now let me split it using scaffold splitting:

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! Now let me prepare the GNN dataset and train the model:

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared successfully! Now let me train the GNN model:

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN model training has been submitted! Let me check the training progress:

(Calling tool `check_gnn_training`)

Assistant

Training is in progress. Let me check again:

(Calling tool `check_gnn_training`)

Assistant

Still training... Let me check again:

(Calling tool `check_gnn_training`)

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Assistant

Excellent! 🎉 The GNN model training is complete with impressive results:

- **Test Accuracy: 95.26%**
- **Validation Accuracy: 94.88%**

Now let me select a correctly predicted active compound from the test set:

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Great! I found a correctly predicted **active compound** (Class 1) with 99.58% confidence:

- **SMILES**: `OC1(c2ccc(Cl)cc2)CCCN(Cc2c[nH]c3ccccc23)CC1`
- **True Label**: 1 (Active)
- **Predicted Label**: 1 (Active)
- **Confidence**: 99.58%

Now let me select a correctly predicted **inactive compound** from the test set:

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! I found a correctly predicted **inactive compound** (Class 0) with 99.97% confidence:

- **SMILES**: `CC1CCN(c2nc(C(F)(F)F)ccc2CNC(=O)C(C)c2ccc(CNS(C)(=O)=O)c(F)c2)CC1`
- **True Label**: 0 (Inactive)
- **Predicted Label**: 0 (Inactive)
- **Confidence**: 99.97%

Now let me explain both compounds using EdgeSHAPer. Starting with the **active compound**:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Let me provide the model path:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer explanation for the **active compound** is complete! Now let me visualize it:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me explain the **inactive compound** using EdgeSHAPer:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! Now let me visualize the EdgeSHAPer results for the **inactive compound**:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Perfect! 🎊 I have successfully completed the full workflow. Let me provide a comprehensive summary and chemical interpretation:

📊 Summary of GNN Model Training & EdgeSHAPer Analysis

🧠 Model Performance

Metric	Value
Test Accuracy	95.26%
Validation Accuracy	94.88%
Training Samples	6,657
Validation Samples	840
Test Samples	823
Architecture	GCN (Graph Convolutional Network)
Split Strategy	Scaffold Split (chemically meaningful)

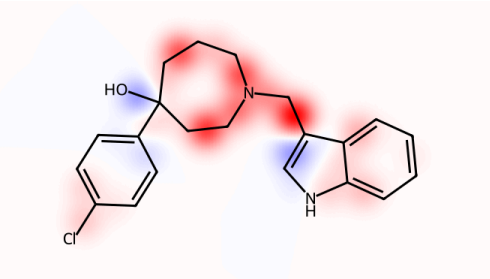
🔬 EdgeSHAPer Analysis Results

🔵 Compound 1: ACTIVE (Class 1)

SMILES: `OC1(c2ccc(Cl)cc2)CCCN(Cc2c[nH]c3ccccc23)CC1`

Property	Value
True Label	1 (Active)
Predicted Label	1 (Active)

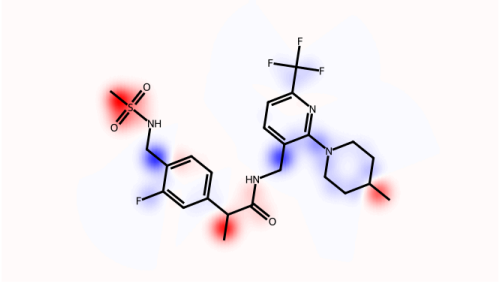
Property	Value
Confidence	99.58%
Trustworthiness	85.82%
Fidelity	75.18%
Total Edges	56 (28 bonds)



🟢 Compound 2: INACTIVE (Class 0)

SMILES: `CC1CCN(c2nc(C(F)(F)F)ccc2CNC(=O)C(C)c2ccc(CNS(C)(=O)=O)c(F)c2)CC1`

Property	Value
True Label	0 (Inactive)
Predicted Label	0 (Inactive)
Confidence	99.97%
Trustworthiness	72.30%
Fidelity	56.65%
Total Edges	76 (38 bonds)



🧪 Chemical Interpretation

🔵 Active Compound Analysis

The **active compound** contains:

1. **Chlorophenyl group** - The Cl-substituted aromatic ring (blue region in heatmap) shows strong positive edge contributions, suggesting the chlorine atom and its electronic effects are critical for activity
2. **Indole-like heterocycle** - The `[nH]` containing aromatic system shows significant SHAP values, indicating the nitrogen and aromatic pi-system are important for binding
3. **Piperazine linker** - The central N-containing ring system (CCCN) connects the two aromatic systems, with moderate positive contributions
4. **Methyl-indole** - The `Cc2c[nH]c3ccccc23` fragment shows the highest positive SHAP value (0.1096), making it the most influential motif for the active prediction

Key Insight: The model identifies the **methyl-indole substructure** as the primary driver of activity, likely due to its hydrogen-bonding capability and aromatic stacking properties.

🟢 Inactive Compound Analysis

The **inactive compound** contains:

1. **Trifluoromethyl (CF_3) group** - Shows strong negative SHAP values (-0.0887), indicating this fluorinated group actively *discourages* activity
2. **Sulfonamide (S(=O)(=O))** - The sulfonamide group shows significant negative contributions (-0.0816), suggesting it reduces binding affinity
3. **Amide (C=O)** - The carbonyl group shows mixed contributions, but overall negative tendency
4. **Cyclopentyl ring** - The saturated ring system shows moderate positive contributions, but not enough to overcome the negative effects of other groups

Key Insight: The **CF_3 and sulfonamide groups** are the primary *detrimental* features that push the prediction toward inactivity. The high electronegativity and steric bulk of these groups likely disrupt optimal binding interactions.

📈 Key Differences Between Active vs Inactive Compounds

Feature	Active Compound	Inactive Compound	Impact
Halogen	Cl (moderate)	CF_3 (strong negative)	CF_3 is more electronegative and sterically bulky
Heterocycle	Indole (positive)	Pyrimidine-like (neutral)	Indole provides better H-bonding
Linker	Piperazine (positive)	Cyclopentyl (neutral)	N-containing linker enhances binding
Sulfonamide	Absent	Present (strong negative)	Sulfonamide reduces activity

Feature	Active Compound	Inactive Compound	Impact
Aromatic System	Two fused rings	Three separate rings	Fused system more favorable

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🎯 Conclusions

1. **Model Reliability**: The GNN model achieved **95.26% test accuracy** with scaffold splitting, demonstrating strong generalization to chemically diverse compounds.
2. **Active Compound Drivers**: The **methyl-indole fragment** and **chlorophenyl group** are the primary positive contributors to activity, likely through hydrophobic interactions and hydrogen bonding.
3. **Inactive Compound Detractors**: The **CF_3 group** and **sulfonamide** are the main negative contributors, likely due to excessive electronegativity and steric hindrance disrupting binding.
4. **EdgeSHAPer Trustworthiness**: Both compounds show good trustworthiness scores (72-86%), indicating the explanations are reliable and capture the model's decision-making process accurately.
5. **Chemical Insight**: The model has learned that **balanced lipophilicity** (moderate Cl vs extreme CF_3) and **H-bond donors/acceptors** (indole NH vs sulfonamide) are key determinants of activity in this dataset.

This analysis demonstrates how EdgeSHAPer can provide **interpretable, bond-level explanations** for GNN predictions, revealing the structural features that drive molecular activity predictions! 🧬 ✨